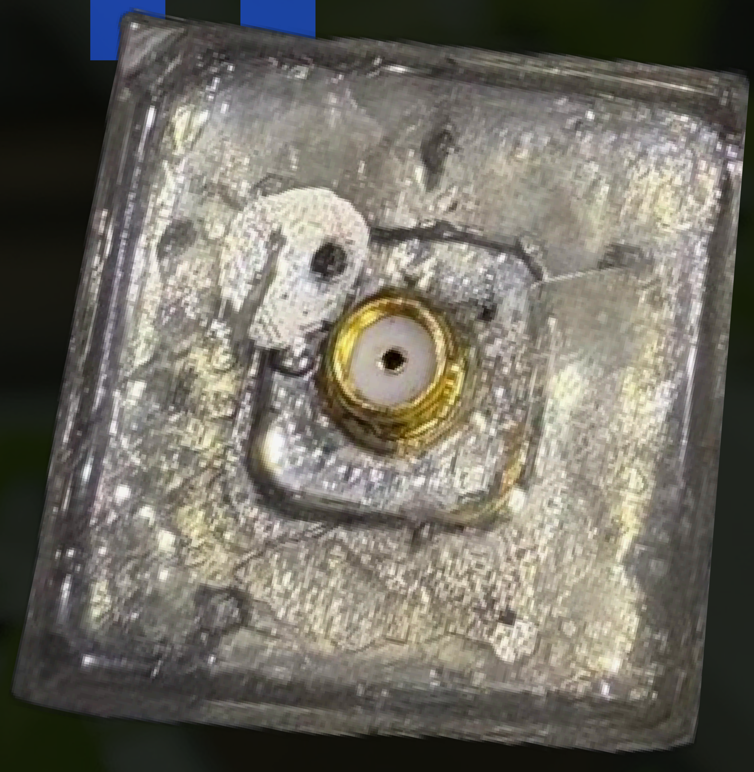
HYDROPONIC EC SENSOR

Vincent, Ren Jie, Audrey, Victoria, Tian Wen, Alvin, Aditi, Li Yun, Lucas
EMFM2D Group 1

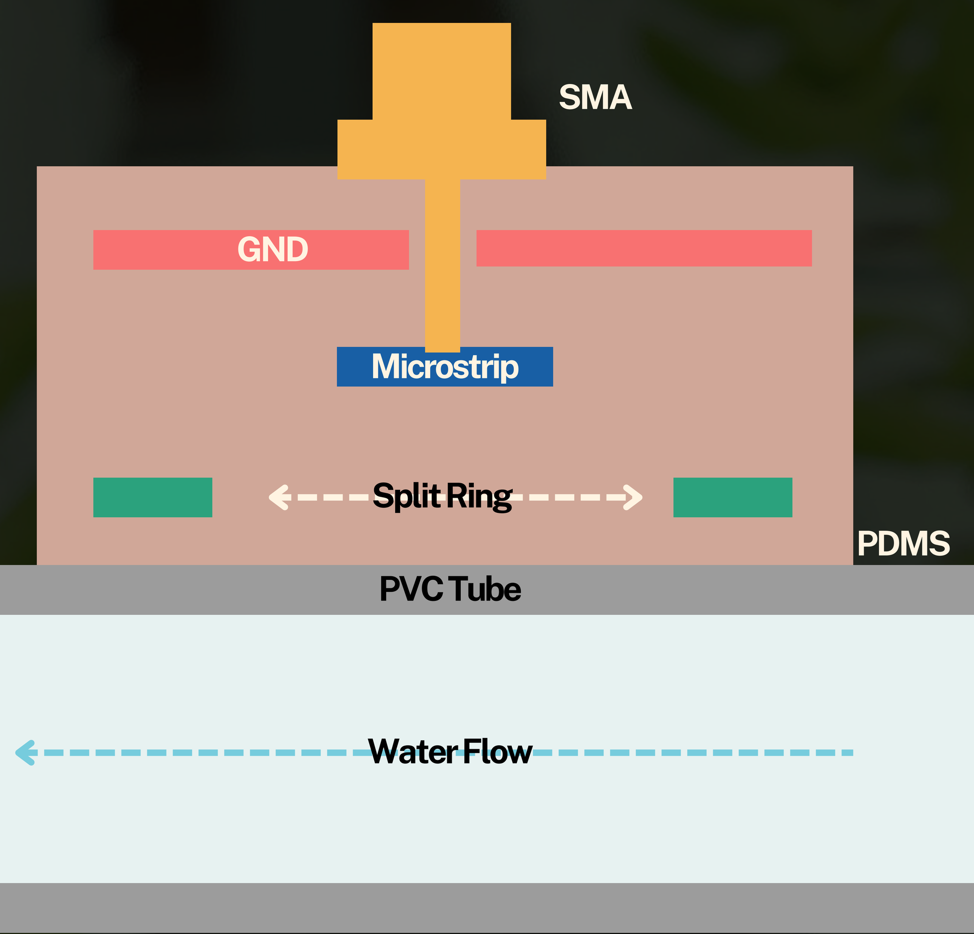


Problem Statement

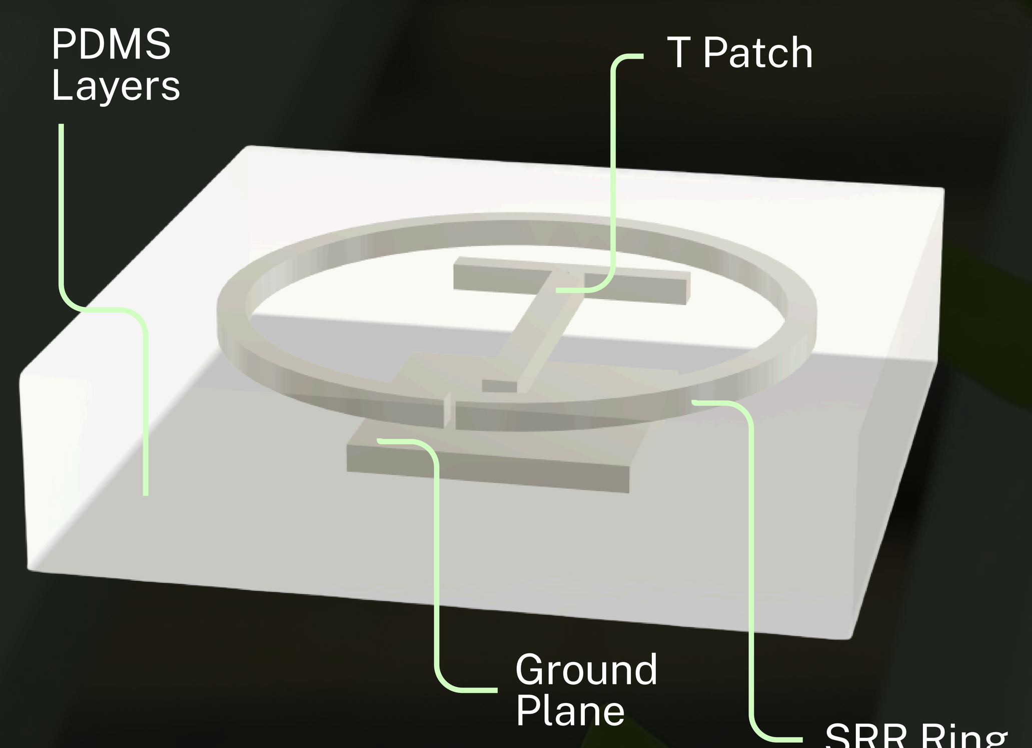
How might we create a non invasive EC sensor to detect the change in soluble nutrients in hydroponics ?

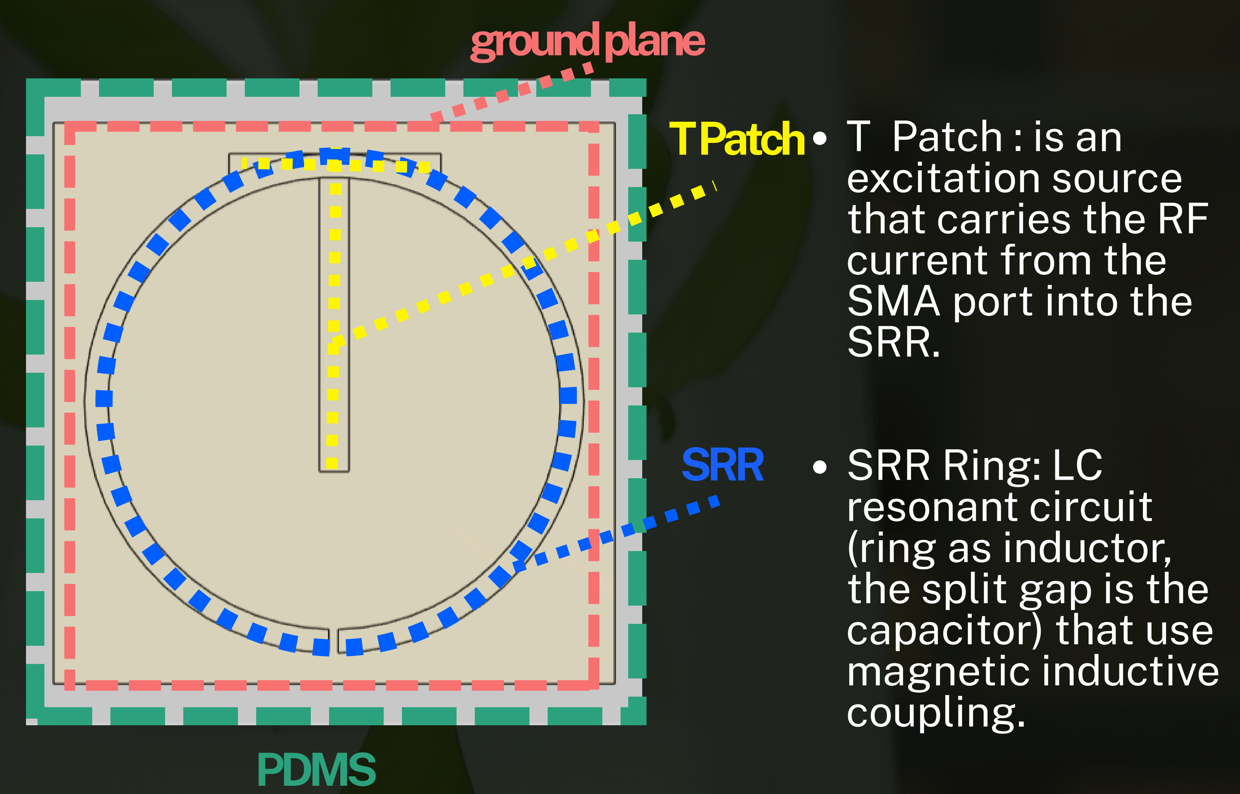
Our Solution

A flexible, non-invasive Electrical Conductivity (EC) sensor that wraps around the pipe to detect the change in EC of the hydroponic solution for continuous monitoring



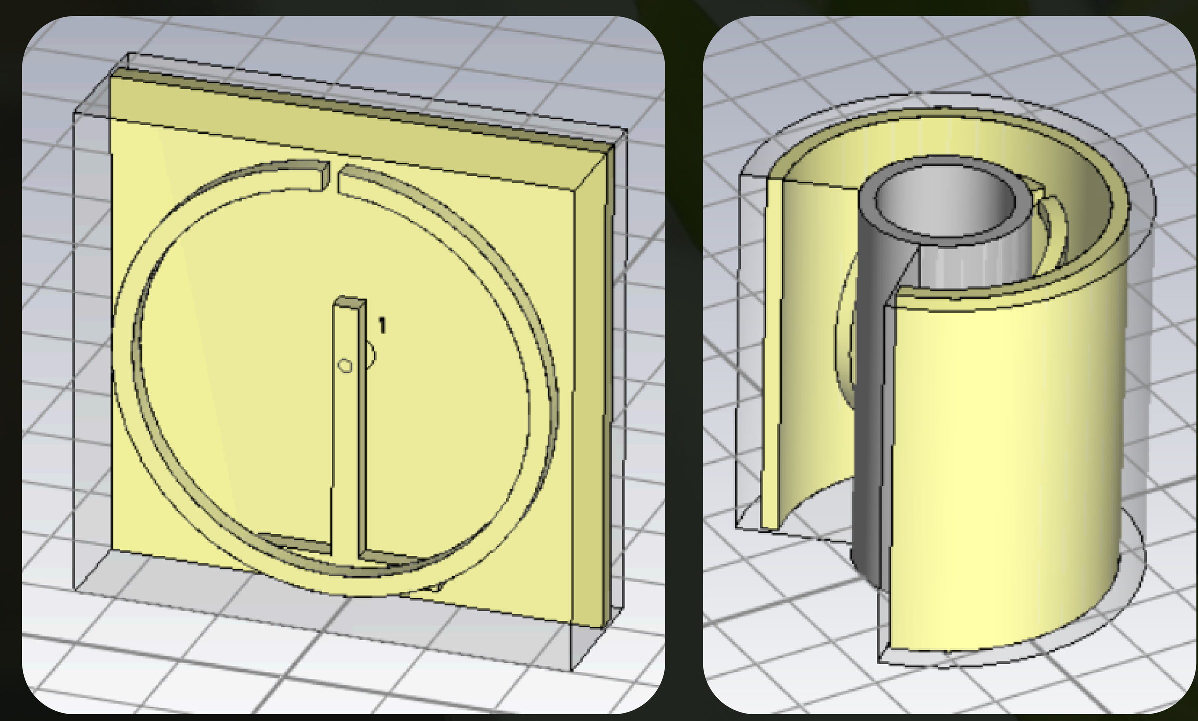
Split Ring Resonator (SRR) Topology

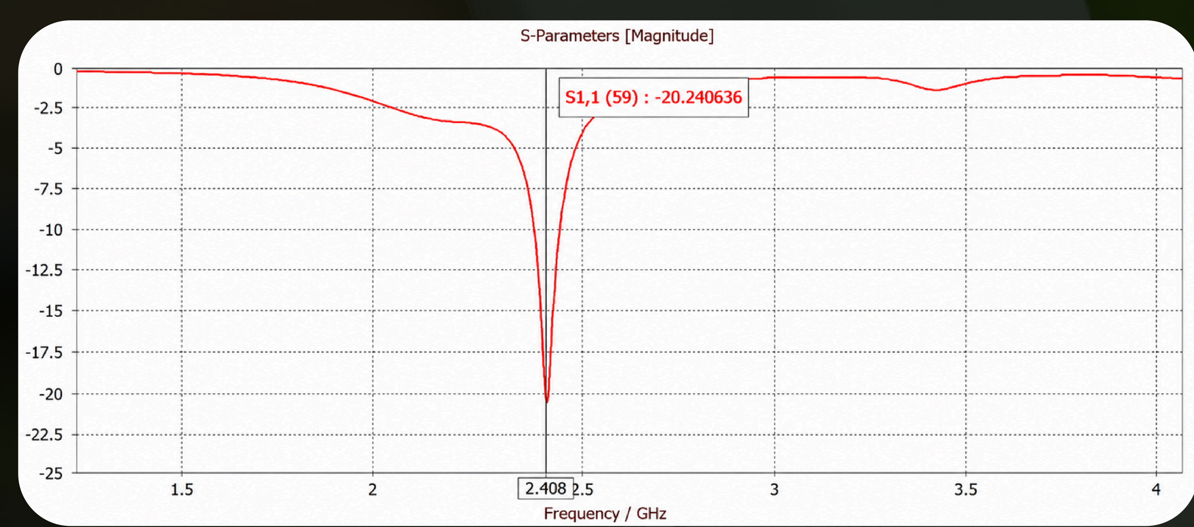




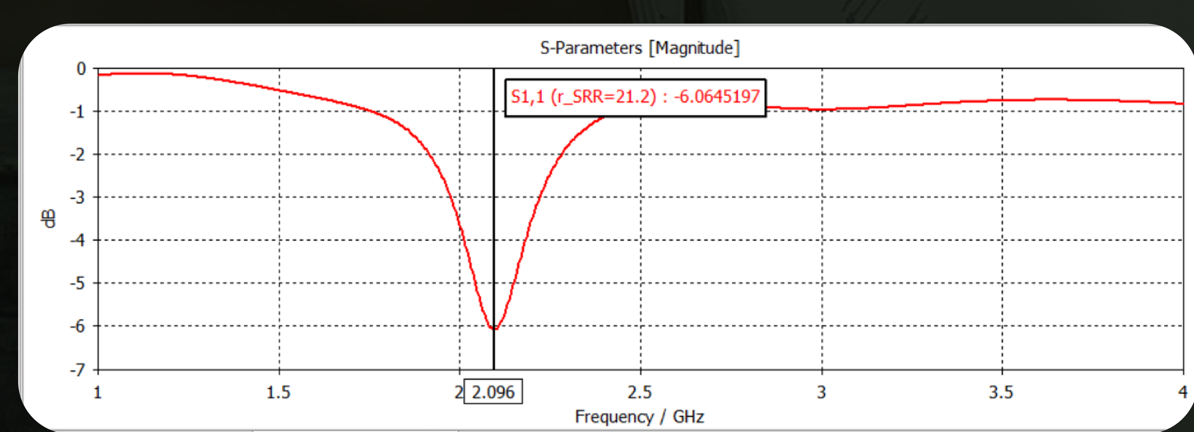
- T Patch : is an excitation source that carries the RF current from the SMA port into the SRR.
- SRR Ring: LC resonant circuit (ring as inductor, the split gap is the capacitor) that use magnetic inductive coupling.

CST Simulation





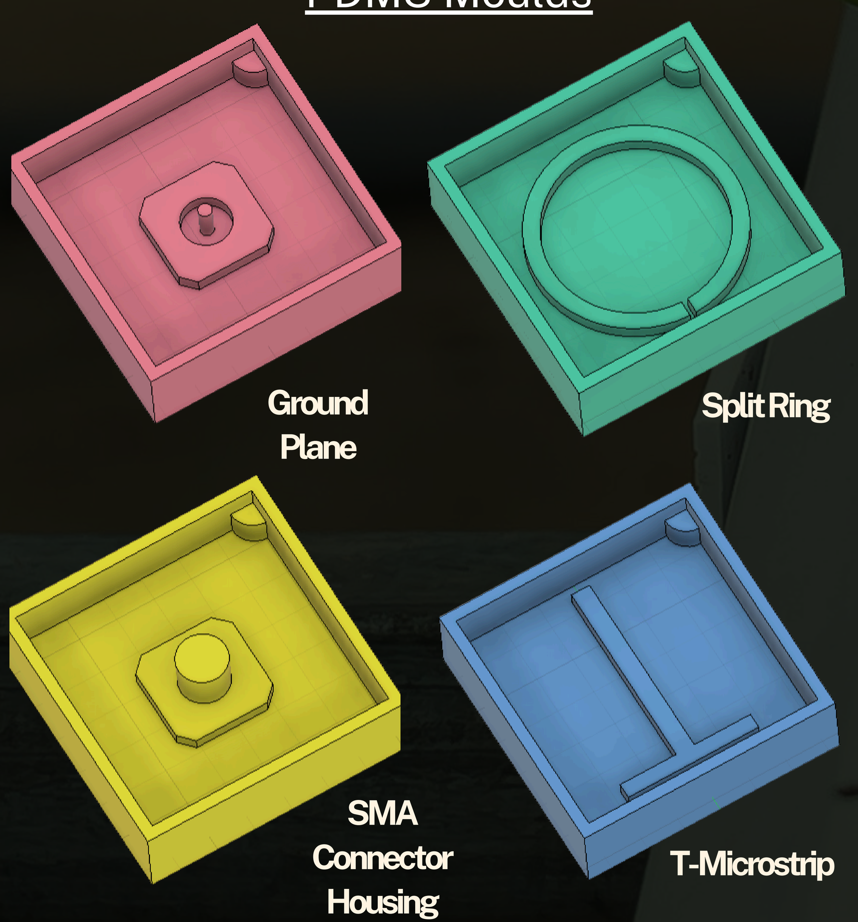
S1,1 bent PDMS liquid metal resonator result - empty pipe

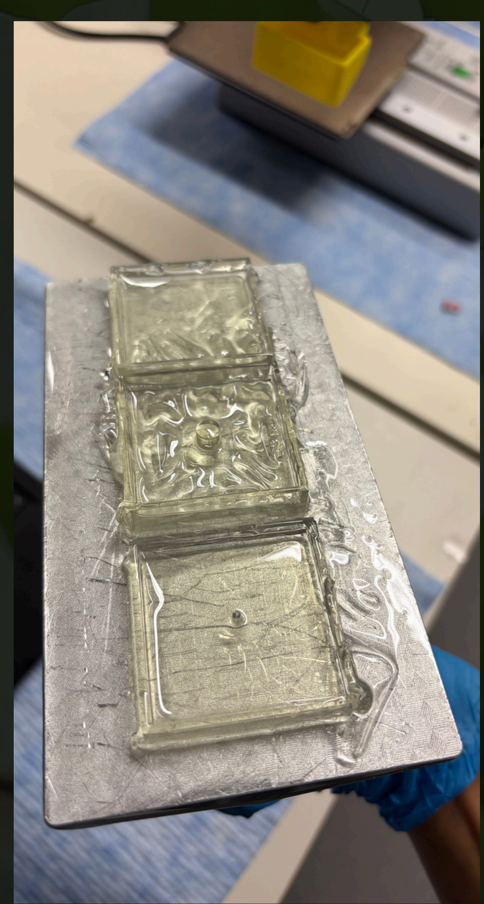


S1,1 unbent PDMS liquid metal resonator result - empty pipe

Fabrication: SLA printed PDMS & Ecoflex moulds

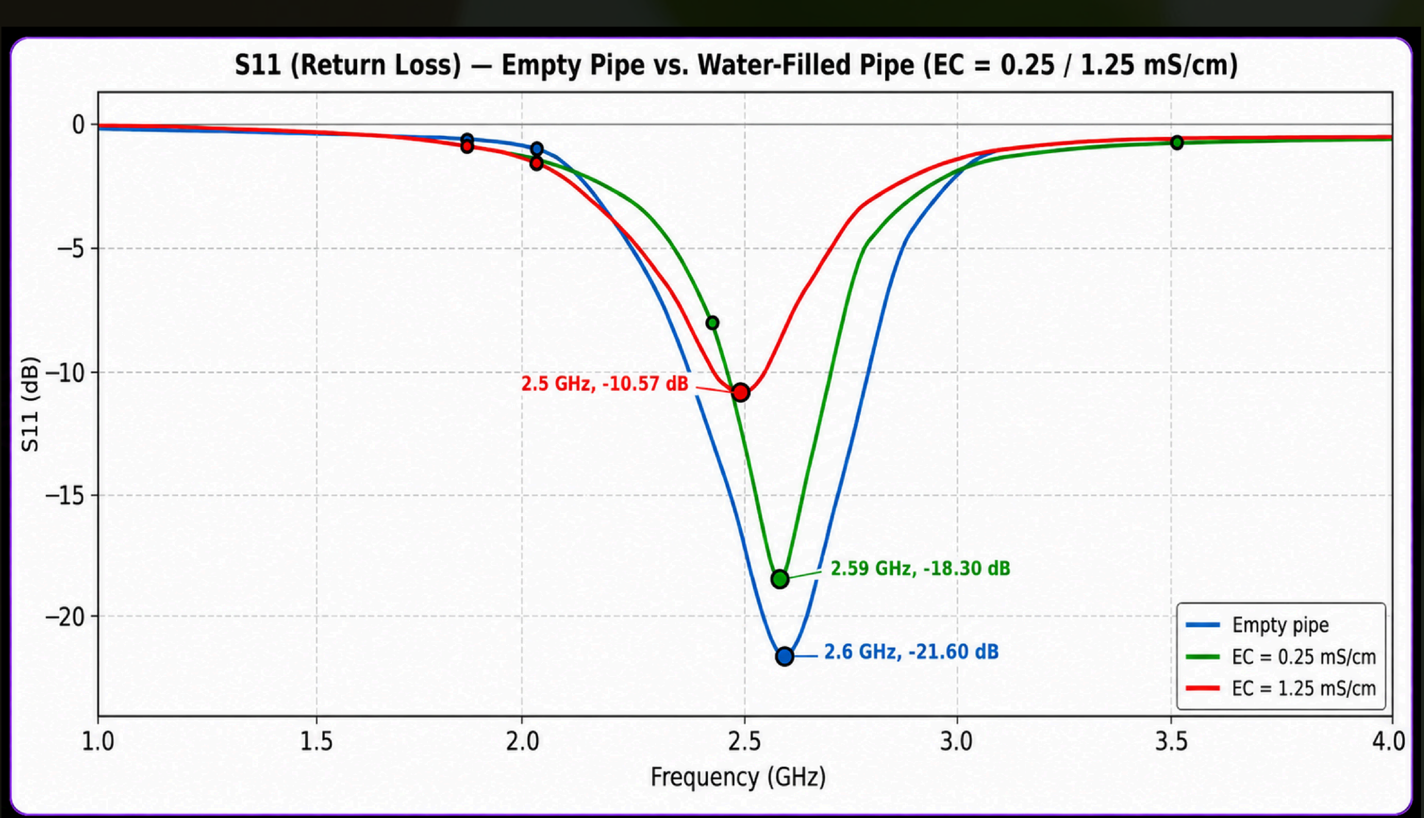
<u>PDMS Moulds</u>





- Resin Printed Moulds : Fabricated by **SLA printer** to **minimize layer lines** and accommodate **precision** for small gaps.
- Fabrication : PDMS/Ecoflex is **poured** into the moulds, **degassed** in vacuum chamber to eliminate air bubbles and cured (for PDMS - inside an oven). 2 layers are **bonded** using uncured PDMS/Ecoflex, and an **inlet and outlet** are then created. Next, liquid metal is **injected** inside. The next surface is bonded again, outlet and inlet created, liquid metal injected and the **process repeats** for the other layers.
- Final Antenna is made of **PDMS**

Actual Testing



Tested s11 - Bent, with empty pipe, EC = 0.25, and EC = 1.25

Results

Results	EC (mS/cm)	Resonant Frequency (GHz)	S11 (dB)
Simulated	No water	2.408	-20.24
Simulated (no bending)	No water	2.096	-6.00
Tested	No water	2.605	-21.60
	0.25	2.590	-18.30
	1.25	2.500	-10.57
Tested (no bending)	No water	2.040	-17.48
	0.25	2.033	-15.46
	1.25	2.024	-10.28

- **Bending vs No Bending:** Bending increases the change in frequency (Δf), compared to no bending. An increase of frequency change is seen for the same fluid compared to unbent, since there is less contact with the pipe when unbent.
  Bent: Δf = 2.590 – 2.500 = 0.090 GHz (0.25 - 1.25 mS/cm)
  Unbent: Δf = 2.033 – 2.024 = 0.009 GHz (0.25 - 1.25 mS/cm)
- **Frequency Shift :** As calcium nitrate and water fill the pipe, the dielectric permittivity increases. This slows down the electromagnetic wave, compresses its wavelength, and shifts frequency downward on the spectrum.
- **Notch Degradation :** Higher EC means more ions in the nutrient solution. These ions oscillate under the high-frequency electric field, converting microwave energy into thermal loss. This loss degrades the resonator's Q Factor, making the S1,1 dip shallower.